to be lean at this stage but the amount of electrolyte in all testing should be controlled the same, for example 75 μl which sufficiently 'floods' the entire coin cell.

If a high CE is achievable in Li (50 μm)/Cu cells under strictly controlled conditions, then an anode-free cell can be utilized to understand the 'worst case' scenario in which the minimum cycling number of a LMB can be estimated in realistic conditions. If a LFP is used as the cathode in the anode-free cell, the value of measured CEs will be close to those obtained in Li/Cu cells due to the relatively narrow electrochemical window of LFP. If a NMC cathode is used with a cut-off voltage of > 4.3 V (versus Li^+/Li) being applied, CE would decrease. The cathode loading in anode-free cells is also suggested to be 4 mAh cm^{-2} to be consistent with Li/Cu cells discussed earlier. One can modify the cathode loading based on the specific need but in either Li/Cu or anode-free cells, the amount of deposited Li and electrolyte content need to be consistent and reasonable. High voltage (> 4.3 V) anode-free cell usually cannot last for a long stable cycling but it provides critical information of the minimum cycling stability that a Li metal cell can operate using the combined cathode and electrolyte.

If CE measured in anode-free cells are still high, the success rate of implementing the functioning materials in realistic high-energy LMBs will be high. A Li (50 μm)/NMC (4 mAh cm^{-2}) coin cell can be assembled to validate the new ideas. Because of the increase of Li source on the anode side, the CE and cycle life of Li/NMC cell will increase compared to completely anode-free cell counterpart. Then the amount of the electrolyte in the Li (50 μm)/NMC (4 mAh cm^{-2}) cells can be gradually reduced to become close to the realistic high energy cells. The electrolyte content in a realistic 300 Wh kg^{-1} LMB[2,3] is about 3 g $(Ah)^{-1}$ which translates to about 13 μL in the Li/NMC coin cell assuming electrolyte density is 1.2 g cm^{-3} and the electrode area is 1.27 cm^2[22]. Note that the coin cell parameters in Fig. 4 are only representative values for constructing high-energy Li metal pouch cells. The exact number can vary but should be within a reasonable range[28].

Through this CE measurement protocol, the results obtained from each different cell configuration can be correlated to each other and provide a clear map to understand the range of stable cycling and evaluate its success in realistic batteries. The same principles can be applied to study Mg, Zn and Na batteries where a metal is used as the anode.

Conclusions

The differences and similarities of CE in Li-ion and rechargeable Li metal batteries have been discussed in this work. CE in different cell configurations such as an anode-free cell, Li/Cu cell and Li/Li symmetrical cells are valuable for evaluating materials but the intrinsic principles governing their CE and cycling life are quite different. In addition to electrochemical window and electrolyte compatibility, the original thickness of Li, type and content of electrolyte and cathode mass loading all need to be considered to fully understanding CE and its relation to LMB lifespan. The artificial inflation of CE in Li metal batteries is a general phenomenon in all rechargeable metal batteries which needs to be considered when elaborating testing results. A protocol for measuring CEs in different coin cell configurations has been suggested to unravel the fundamental interrelations of those cells and apply CEs to estimate the range of cycle life for realistic high-energy LMBs. The fundamental understanding of this work is broadly applicable in metal-based battery technologies such as Mg, Zn and Na batteries.

Methods

Cell preparation. Single-side coated electrodes of $LiNi_{0.8}Mn_{0.1}Co_{0.1}O_2$ (NMC811) and $LiNi_{0.6}Mn_{0.2}Co_{0.2}O_2$ (NMC622) (area capacity: 4 mAh cm^{-2}) used as cathode in coin cell were provided by Advanced Battery Facility (ABF) at Pacific Northwest National Laboratory (PNNL). 10 μm Cu foil and 50 μm Li metal foil attached on 10 μm Cu foil were used as anodes in different cell configurations. Polyethylene (PE) film was used as a separator in coin cell. The ingredients of electrolytes such as lithium hexafluorophosphate ($LiPF_6$), ethylene carbonate (EC), ethyl methyl carbonate (EMC), vinlyene carbonate (VC) (BASF Battery Materials), lithium bifluorosulfnyl imde (LiFSI) (Nippon Shokubai), triethylphosphate (TEP) (Sigma Aldrich, ReagentPlus, >99.8%) and bis(2,2,2-trifluoroethyl) ether (BTFE) (SynQuest Labs, 99%) were used as received. The conventional electrolyte with the formula of 1.0 M $LiPF_6$ in EC/EMC (3:7 by weight) + 2 wt% VC and the compatible electrolyte with the formula of 1.2 M LiFSI in TEP/BTFE (a molar ratio of LiFSI:TEP:BTFE at 0.75:1:3) were prepared in argon-filled glovebox (MBraun), where both oxygen and moisture levels were below 0.1 ppm.

The electrochemical performances of all coin cells were evaluated in 2032-type coin cells (National Research Council Canada) on Land battery testers and the cells were filled with 75 μL electrolytes unless otherwise specified (such as 15 μL). 1C rate corresponded to a current density of 4.0 mA cm^{-2}. All Li metal and anode-free/NMC cells, after having two formation cycles at C/10, were cycled by constant current charging at C/10 rate to 4.4 V and then constant voltage charging at 4.4 V to C/20, after that, constant current discharging the cells at C/3 rate to 2.7 V at room temperature in a TestEquity temperature chamber. For the Li/Cu cell, the Li depositing amount on Cu is 4 mAh cm^{-2} and the cut-off voltage of stripping from Cu is 1.0 V. After two formation cycles at current density of 0.4 mA cm^{-2}, the coin cells were cycled by depositing at current density of 0.4 mA cm^{-2} and stripping at current density of 1.32 mA cm^{-2}.

Characterization. For sample preparation, all cycled electrodes were obtained by disassembling the cycled coin cells inside a glovebox and were immediately rinsed with pure EMC to remove the residual electrolyte, and then dried under vacuum. Samples were hermetically sealed inside an air-tight bottle for transferring. Scanning electron microscope (SEM) images were collected on JSM-IT200 (JEOL, Japan) equipped with EDS.

Data availability

All relevant data are included in the paper.

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Competing interests

The authors declare no competing interests.

Additional information

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